Cisco PE Router BGP Route Churn, OSPF Adjacency Loss, and ISIS LSP Flooding Troubleshooting Procedure for MG-PE-04 at MG Road Core

Node: MG-PE-04 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_3610

Description

The MG-PE-04 node, situated at the MG Road Core site, functions as a Provider Edge (PE) router within the telecom network, leveraging Cisco's robust routing platform. The PE role is critical for interconnecting customer networks to the provider's backbone. The failure domain of Routing Protocol Flap, as indicated by alarm code ALARM_3610 with a critical severity, suggests instability in the routing protocols. This could stem from BGP route churn, where the Border Gateway Protocol (BGP) experiences fluctuations in route advertisements, leading to network instability. Similarly, OSPF (Open Shortest Path First) adjacency loss implies a disruption in the link-state advertisements between neighboring routers, potentially causing routing loops or black holes. ISIS (Intermediate System to Intermediate System) LSP flooding refers to the excessive propagation of Link-State Packets (LSPs) within the ISIS domain, which can overwhelm the network and lead to routing instability. These issues can arise from various factors, including misconfigurations, hardware failures, or external network changes. Given the complex interplay between these routing protocols and the critical role of the PE router, a meticulous troubleshooting approach is essential to identify and rectify the root cause without disrupting the network services.

Pre-requisite Checks

- Verify the physical integrity and power status of MG-PE-04 at the MG Road Core site.
- Check the status of redundant links and ensure that they are operational.
- Confirm that all necessary tools and access credentials are available for the troubleshooting process.

Potential Root Causes

Cause	Probability
BGP Route Churn due to Incorrect BGP Configuration	High
OSPF Adjacency Loss due to Interface Flaps	Medium
ISIS LSP Flooding due to Network Topology Changes	Low

Diagnostic Commands & Expected Outputs

```
Command: show ip bgp, Expected Output: Displays the current BGP routing table, allowing for the identification of any inconsistent or flapping routes., Description: This command isolates issues in the BGP control plane by examining the BGP routing table for any signs of route churn or instability.
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```
Command: show ip ospf neighbor, Expected Output: Lists all OSPF neighbors and their current state, helping to identify any adjacency issues., Description: This command is used to diagnose OSPF adjacency problems by examining the state of neighboring routers and identifying any that are not in a full state.

Command: show isis database, Expected Output: Displays the ISIS link-state database, useful for identifying any LSP flooding issues., Description: This command examines the ISIS database for any excessive LSPs that could indicate flooding, which might be causing routing instability.
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Resolution Steps

Step 1: Step: Adjust BGP Configuration, Command: router bgp neighbor route-map out, Description: Apply a route-map to filter out flapping routes from being advertised to neighbors, thus stabilizing the BGP routing table.

Step 2: Step: Stabilize OSPF Adjacencies, Command: interface ip ospf hello-interval , Description: Adjust the OSPF hello interval on unstable interfaces to reduce the likelihood of adjacency loss due to transient interface issues.

Step 3: Step: Mitigate ISIS LSP Flooding, Command: router isis lsp-refresh-interval , Description: Increase the LSP refresh interval to reduce the frequency of LSP flooding, thus alleviating potential routing instability.

Post-Check Verification

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Command: show ip bgp, Expected Output: A stable BGP routing table with no signs of route churn., Description: Verify that BGP route churn has been resolved and the routing table is stable.

Command: show ip ospf neighbor, Expected Output: All OSPF neighbors are in a full state., Description: Confirm that OSPF adjacencies are stable and all neighbors are fully connected.

Command: show isis database, Expected Output: The ISIS database shows no signs of LSP flooding., Description: Verify that ISIS LSP flooding has been mitigated and the database is stable.
```

Roll-back Procedures

- Step: Revert BGP Configuration, Command: no router bgp neighbor route-map out, Description: Remove the applied route-map to revert BGP configuration to its pre-change state if the fix fails.

- Step: Reset OSPF Hello Interval, Command: interface no ip ospf hello-interval , Description: Revert the OSPF hello interval to its default or previous setting if adjusting it does not resolve the adjacency issue.

- Step: Revert ISIS LSP Refresh Interval, Command: router isis no lsp-refresh-interval , Description: Revert the LSP refresh interval to its default setting if changing it does not mitigate LSP flooding.